

Summer Workshop: Causal Inference for Uber Pool

From observed operational data to a causal decision model

Business Case and Causal Question

Uber tested a policy change in Uber Pool: allowing riders to wait up to **5 minutes** instead of **2 minutes** before matching. The operational intuition is that a longer waiting window may improve shared matching, increase the number of pooled rides, and reduce matching costs. However, the policy may also increase rider cancellations or change the type of trips completed. The central question is not predictive. The goal is not only to ask whether treated periods have higher or lower payouts. The causal question is:

What is the causal effect of increasing the Uber Pool waiting window from 2 minutes to 5 minutes on operational performance?

The main estimand is the average treatment effect:

$$ATE = \mathbb{E}[Y(1) - Y(0)],$$

where $Y(1)$ is the outcome under the 5-minute waiting policy and $Y(0)$ is the outcome under the 2-minute waiting policy.

Learning Objectives

By the end of the workshop, each team should be able to:

- distinguish association, correlation, and causality in the Uber Pool setting;
- identify the treatment, outcome, contextual variables, mediators, and possible post-treatment variables;
- construct a causal DAG that matches a specific business question;
- explain why adding more controls can improve prediction but damage causal interpretation;
- propose a defensible causal model and communicate it as a decision recommendation.

Data Dictionary

Workshop Tasks

Part 1. Define the causal problem

Answer the following questions before drawing any model:

1. What is the intervention? Write it as a **do**-operation: $do(X = x)$.
2. What is the main outcome? Is `total_driver_payout` a benefit, a cost, or an operational performance measure?
3. What is the main business decision: keep the 5-minute policy, return to 2 minutes, or use it only in specific contexts?
4. Which variables are observed before or at the time of treatment, and which variables may happen after treatment?

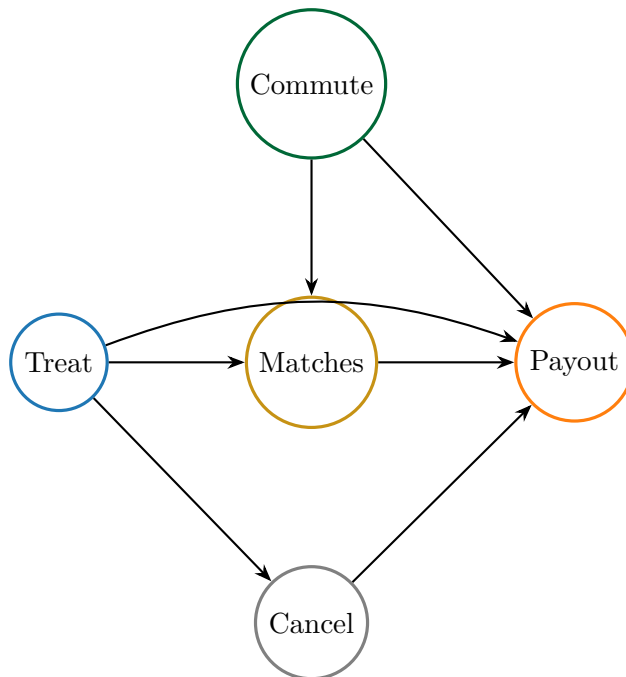
Variable	Type	Definition
period_start	Time	Time period in which the data were gathered.
wait_time	String	Matching policy: 2 mins if riders could wait up to 2 minutes; 5 mins if riders could wait up to 5 minutes.
treat	Boolean	Treatment indicator: TRUE when wait_time = 5 mins; FALSE when wait_time = 2 mins.
commute	Boolean	Rush-hour indicator: TRUE during commuting periods, approximately 7:00–9:40 AM or 3:00–5:40 PM; FALSE otherwise.
trips_pool	Numeric	Total number of completed Pool trips in the current period. Each matched ride request is counted as a separate trip.
trips_express	Numeric	Total number of completed Express Pool trips in the current period. Each matched ride request is counted as a separate trip.
rider_cancellations	Numeric	Total number of requested trips cancelled by riders in the current period.
total_driver_payout	Numeric	Total pesos paid to drivers for completed trips in the current period; this represents Uber's total matching cost.
total_matches	Numeric	Number of completed trips paired with at least one other rider for part of the trip. Each matched ride request is counted separately.
total_double_matches	Numeric	Number of completed trips paired with at least two other riders for part of the trip. Each matched ride request is counted separately.

Part 2. Build a first causal DAG

Construct a DAG that includes at least the following nodes:

treat, commute, total_matches, rider_cancellations, total_driver_payout.

A suggested starting structure is:



Discuss whether `commute` is a context variable, a confounder, an effect modifier, or all of these. Then decide whether `total_matches` and `rider_cancellations` are mediators, outcomes, colliders, or variables that should be handled carefully.

Part 3. Compare three causal models

Your team must compare the meaning of three candidate models. The goal is not to choose the largest model, but to explain what causal question each model answers.

1. Naive association model

$$\text{total_driver_payout}_i = \beta_0 + \beta_1 \text{treat}_i + \varepsilon_i.$$

What can this model say? What can it not say?

2. Adjusted total-effect model

$$\text{total_driver_payout}_i = \beta_0 + \beta_1 \text{treat}_i + \beta_2 \text{commute}_i + \varepsilon_i.$$

Under what assumptions can β_1 be interpreted as a total policy effect?

3. Mechanism or direct-effect model

$$\text{total_driver_payout}_i = \beta_0 + \beta_1 \text{treat}_i + \beta_2 \text{commute}_i + \beta_3 \text{total_matches}_i + \varepsilon_i.$$

Why does adding `total_matches` change the question from “What is the total effect?” to “What is the direct effect after holding matching fixed?”

Part 4. Identify risks of causal misinterpretation

Prepare one clear explanation of each risk:

- **Confounding:** a third variable affects both treatment assignment or exposure and the outcome.
- **Mediation:** part of the treatment effect flows through variables such as `total_matches` or `rider_cancellations`.
- **Post-treatment bias:** controlling for a variable affected by treatment may block part of the effect.
- **Collider bias:** conditioning on a variable jointly influenced by treatment and user response may create a spurious relationship.

Final Deliverable: 7-Minute Presentation

Each group must prepare a short presentation of **maximum 7 minutes**. There is **no written report**. The presentation should include no more than five slides:

1. **Business question:** What decision should Uber make about the 5-minute Pool waiting policy?
2. **Data and variables:** Identify treatment, outcome, context variables, mediators, and possible post-treatment variables.
3. **DAG:** Present your causal graph and explain the main pathways.
4. **Model choice:** Compare the naive, adjusted, and mechanism models. State which one answers your main question.
5. **Decision recommendation:** Should Uber scale, reject, or conditionally apply the policy? What evidence would you still need?